

NileCAM81_CUOAGX

Developer Guide



e-con Systems

Think Camera. Think e-con.

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Introduction to NileCAM81_CUOAGX

NileCAM81_CUOAGX is family of NileCAM cameras from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson™ AGX Xavier and NVIDIA® Jetson AGX Orin™ development kits. NileCAM81_CUOAGX is an 8 MP GMSL camera that contains a 1/1.7"AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to Serializer board through a rugged FAKRA connector. NileCAM81_CUOAGX can be directly interfaced with Jetson AGX Xavier™ /Orin™ development kit through J509 connector.

The NVIDIA® Jetson™ AGX Xavier/Orin development kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

e-con Systems also provides ecam_tk1_guvcview sample application that demonstrates the features of this camera. However, this camera can utilize any Video for Linux version 2 (V4L2) application.

The commands and output messages in this manual are represented by different colors as shown in below table.

Table 1: Notation of Colors

Color	Notation
Blue	Commands running in host PC
Red	Output message in host PC
Green	Commands running in Jetson development kit
Orange	Output messages in Jetson development kit

This document explains how to setup the Jetson AGX Orin™/Xavier™ development kit for using NileCAM81_CUOAGX camera.

Software Requirements

The software requirements are as follows:

- Cross compiler toolchain
- Linux for Tegra (L4T) release package and sample root filesystem (rootfs)

Prerequisites

This section describes the requirements to use NileCAM81_CUOAGX on the Jetson AGX Orin™/Xavier™ development kit.

The prerequisites are as follows:

- Host PC which runs Ubuntu 20.04 (64-bit).
- NVIDIA® provided L4T release and corresponding sample rootfs for Jetson AGX Orin™/Xavier™ development kit.
- A kernel image and device tree blob (DTB) file. The release package contains a kernel binary (Image) and the DTB files, which you can download and rebuild from source.
- Module drivers for the NileCAM81_CUOAGX camera.
- A USB cable (Type-C port) to plug into the recovery port of the Jetson AGX Orin™/Xavier™ development kit.

Note: USB-C port located on the other side of the board from the power port is used for flashing the board.

Please refer to the *NileCAM81_CUOAGX_Release_Package_Manifest.pdf* to know the contents of release package and their description.

Setting up Environment

The steps to setup the environment are as follows:

1. Run the following commands to setup the required environment variables.

```
mkdir top_dir/kernel_out -p
mkdir top_dir/kernel_sources
mkdir top_dir/tool_chain
export TOP_DIR=<absolute_path_to>/top_dir
export RELEASE_PACK_DIR=$TOP_DIR/e-CAM_YUV-GMSL-
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_
version>
export L4T_DIR=$TOP_DIR/Linux_for_Tegra
export LDK_ROOTFS_DIR=$TOP_DIR/Linux_for_Tegra/rootfs
export ARCH=arm64
export CROSS_COMPILE=aarch64-buildroot-linux-gnu-
export CROSS32CC=arm-linux-gnueabi-hf-gcc
export TEGRA_KERNEL_OUT=$TOP_DIR/kernel_out
export TOOL_CHAIN=$TOP_DIR/tool_chain
```

```
export
NVIDIA_SRC=$TOP_DIR/kernel_sources/Linux_for_Tegra/source/public
export DISPLAY_SOURCE_DIR=$NVIDIA_SRC/NVIDIA-kernel-module-source-TempVersion
```

2. Run the following command to copy the release package tar file to the staging directory.

```
mv <location_of>/e-CAM_YUV-GMSL-PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_version>.tar.xz $TOP_DIR
```

Downloading Requirements

For building the kernel, a cross compiler toolchain and other tools necessary for compiling are required. You need to download the cross-compiler toolchain from NVIDIA® website.

The steps to download the requirements for building the kernel are as follows:

1. Download the required toolchain from NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download the package and to export the PATH environment for building kernel source.

- a. Download the packages from the NVIDIA® website as listed in below table.

Table 2: GCC Toolchain Package

S.No	Title	Version	Download Link
1	Bootlin Toolchain	9.3	https://developer.nvidia.com/embedded/jetson-linux/bootlin-toolchain-gcc-93

- b. Run the following commands to extract the package in host PC.

```
cd $TOOL_CHAIN
tar -xf $HOME/Downloads/aarch64--glibc--stable-final.tar.gz
```

- c. Run the following command to export PATH environment for building kernel source.

```
export PATH=$TOOL_CHAIN/bin:${PATH}
```

2. Run the following commands to download the required package for extracting sources.

```
sudo apt-get update
sudo apt-get install bison
```

```
sudo apt-get install qemu-user-static
sudo apt-get install build-essential
sudo apt-get install bc
sudo apt-get install lbzip2
sudo apt-get install python
sudo apt-get install flex
sudo apt-get install openssl
sudo apt-get install libssl-dev
sudo apt-get install make
```

3. Download the required L4T release package and sample rootfs from NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download the packages and to copy the downloaded packages are as follows:

- a. Download the packages from the NVIDIA® website as listed in below table.

Table 3: Packages for Jetson AGX Xavier

S.NO	Title	Version	Download Link
1	L4T Jetson AGX Xavier™ and Orin™ driver Package	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/jetson_linux_r35.4.1_aarch64.tbz2/
2	L4T Jetson AGX Xavier™ and Orin™ sample Rootfs	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/release/tegra_linux_sample-root-filesystem_r35.4.1_aarch64.tbz2/

- b. Run the following commands to copy the downloaded package to staging directory.

```
cp
$HOME/Downloads/Jetson_Linux_R35.4.1_aarch64.tbz2
$TOP_DIR

cp $HOME/Downloads/Tegra_Linux_Sample-Root-
Filesystem_R35.4.1_aarch64.tbz2 $TOP_DIR
```

Extracting and Preparing L4T

The steps for extracting and preparing L4T must be performed in host PC as follows:

1. Run the following commands to extract the downloaded L4T release package to navigate a folder with the name Linux_for_Tegra.


```
cd $TOP_DIR  
sudo tar -xjpf Jetson_Linux_R35.4.1_aarch64.tbz2
```

2. Run the following commands to extract the sample rootfs to the rootfs directory which is present inside the Linux_for_Tegra directory.

```
cd $LDK_ROOTFS_DIR  
sudo tar -xjpf $TOP_DIR/Tegra_Linux_Sample-Root-  
Filesystem_R35.4.1_aarch64.tbz2
```

3. Run the following commands to set the package to be ready to flash binaries.

```
cd $L4T_DIR  
sudo ./apply_binaries.sh
```

Extracting the Release Package

Run the following commands to extract the NileCAM81_CUOAGX release package.

```
cd $TOP_DIR  
tar -xaf e-CAM_YUV-GMSL-  
PRODUCTS__JETSON_<L4T_version>_<release_date>_<release_ve  
rsion>.tar.xz
```

To know more about the release package, please refer to the *NileCAM81_CUOAGX-Release_Package_Manifest.pdf*.

Please refer to the *Installation Procedure* section to build kernel with support for NileCAM81_CUOAGX. The procedure would require flashing the eMMC of the Jetson AGX Orin™/Xavier™ development kit for erasing the pre-existing contents.

Installation Procedure

This section describes the steps for building and installing the kernel.

Building from Source

You can use a patch file provided by e-con Systems to use your own kernel image binary and modules along with the NileCAM81_CUOAGX camera on the Jetson AGX Orin™/Xavier™ development kit to build the kernel.

Downloading and Configuring the Kernel

This section describes how you can download and configure the kernel for Jetson AGX Orin™/Xavier™ development kit.

1. Download the kernel source code for L4T from the NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download and configure the kernel for Jetson AGX Orin™/Xavier™ development kit are as follows:

2. Download the packages from the NVIDIA® website as listed in below table.

Table 4: Packages for Jetson AGX Xavier

Title	Version	Download Link
L4T Sources	35.4.1	https://developer.nvidia.com/downloads/embedded/l4t/r35_release_v4.1/sources/public_sources.tbz2/

3. Run the following command to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/public_sources.tbz2
$TOP_DIR/kernel_sources
```

4. Run the following commands to extract the downloaded kernel source code to any path on the host Linux PC.

```
cd $TOP_DIR/kernel_sources
tar -xjf public_sources.tbz2
cd $NVIDIA_SRC
```

5. Run the following commands to extract the kernel source code.

```
tar -xpf kernel_src.tbz2
tar -xjpf nvidia_kernel_display_driver_source.tbz2
```

6. Run the following commands to make sure that both patch command is applied properly in the kernel source.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_AGX_XAVIER_ORIN_L4T35.4.1_kernel.patch
--dry-run

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_AGX_XAVIER_ORIN_L4T35.4.1_module.patch
--dry-run

patch -p1 -i
$RELEASE_PACK_DIR/Kernel/STURDeCAM81_CUOAGX_JETSON_AGX
_XAVIER_ORIN_L4T35.4.1_dtb.patch --dry-run
```

7. Run the following commands to apply the patch file to the kernel source code, if there is no error from dry-run command.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_AGX_XAVIER_ORIN_L4T35.4.1_kernel.patch

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_AGX_XAVIER_ORIN_L4T35.4.1_module.patch

patch -p1 -i
$RELEASE_PACK_DIR/Kernel/STURDeCAM81_CUOAGX_JETSON_AGX
_XAVIER_ORIN_L4T35.4.1_dtb.patch
```

Building and Installing the Kernel

The steps to build and install the kernel are as follows:

1. Run the following commands to build and install the kernel image and modules to the Jetson AGX Orin™/Xavier™ development kit.

```
cd kernel/kernel-5.10/
make O=$TEGRA_KERNEL_OUT defconfig
make O=$TEGRA_KERNEL_OUT Image -j4
make O=$TEGRA_KERNEL_OUT modules -j4
make O=$TEGRA_KERNEL_OUT dtbs
sudo ARCH=arm64 make O=$TEGRA_KERNEL_OUT
modules_install INSTALL_MOD_PATH=$LDK_ROOTFS_DIR
```

2. Run the following commands to build and install the NileCAM81_CUOAGX driver code.

```
cd $NVIDIA_SRC/e-CAM_YUV_GMSL
make ARCH=arm64 KERNEL_PATH=$TEGRA_KERNEL_OUT ar0230
sudo make -C $TEGRA_KERNEL_OUT M=$PWD
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
```

3. Run the following commands to build and install the display driver support for Jetson AGX Orin™/Xavier™ development kit.

```
cd $DISPLAY_SOURCE_DIR
make -C $PWD modules -j $(nproc) TARGET_ARCH=aarch64
ARCH=arm64 CC="$CROSS_COMPILE}gcc"
```

```
LD="${CROSS_COMPILE}ld" AR="${CROSS_COMPILE}ar"
CXX="${CROSS_COMPILE}g++"
OBJCOPY="${CROSS_COMPILE}objcopy"
SYSOUT="$TEGRA_KERNEL_OUT"
SYSSRC="$NVIDIA_SRC/kernel/kernel-5.10"

sudo make M=$PWD/kernel-open -C $TEGRA_KERNEL_OUT
INSTALL_MOD_PATH=$LDK_ROOTFS_DIR modules_install
```

8. Run the following commands to copy the kernel and dtb file to Linux_for_Tegra (L4T_DIR) flashing path.

```
sudo cp $TEGRA_KERNEL_OUT/arch/arm64/boot/Image
$L4T_DIR/kernel/ -f
```

For Jetson AGX Xavier™,

```
sudo cp
$TEGRA_KERNEL_OUT/arch/arm64/boot/dts/nvidia/tegra194-
p2888-0001-p2822-0000-nilecam81_two_lane.dtb
$L4T_DIR/kernel/dtb/tegra194-p2888-0001-p2822-0000.dtb
-f// FOR JETSON AGX ORIN
```

For Jetson AGX Orin™ 32 GB Development Kit,

```
sudo cp
$TEGRA_KERNEL_OUT/arch/arm64/boot/dts/nvidia/tegra234-
p3701-0000-p3737-0000-nilecam81_two_lane.dtb
$L4T_DIR/kernel/dtb/tegra234-p3701-0000-p3737-0000.dtb
-f
```

For Jetson AGX Orin™ 64 GB Development Kit,

```
sudo cp
$TEGRA_KERNEL_OUT/arch/arm64/boot/dts/nvidia/tegra234-
p3701-0005-p3737-0000-nilecam81_two_lane.dtb
$L4T_DIR/kernel/dtb/tegra234-p3701-0005-p3737-0000.dtb
-f
```

Modifying the Rootfs

Run the following commands to modify additional files in the rootfs for the proper functioning of the NileCAM81_CUOAGX camera on the Jetson AGX™ Xavier™ development kit.

```
sudo cp $RELEASE_PACK_DIR/misc/modules
$LDK_ROOTFS_DIR/etc/modules -f

sudo cp $RELEASE_PACK_DIR/misc/xorg.conf.t194_ref
$LDK_ROOTFS_DIR/etc/X11/xorg.conf.t194_ref -f
```

Copying the MCU firmware to Rootfs

Run the following command to copy the mcu firmware files to the rootfs for the proper functioning of the STURDeCAM20/21/25/81_CUOAGX camera on the Jetson AGX Xavier™/Orin™ development kit.

For Jetson AGX Xavier™/Orin™,

```
sudo cp $RELEASE_PACK_DIR/firmware/ar0821_mcu_fw.bin  
$LDK_ROOTFS_DIR/usr/lib/firmware/
```

Flashing the Jetson AGX Xavier/ Jetson Orin Development Kit

The steps to flash the Jetson AGX Xavier™/ Orin™ development kit are as follows:

1. Connect the USB Type-C cable to the host PC and the USB-C port of Jetson AGX Xavier™ or USB Type-C port of Jetson AGX Orin™ development kit.

The location of USB-C port on the Jetson AGX Xavier™ development kit is shown in below figure.



Figure 1: Location of USB-C Port on Jetson AGX Xavier Development Kit

Note: USB-C port located on the other side of the board from the power port is used for flashing the board.

The location of USB-B port on the Jetson Orin™ development kit is shown in below figure.



Figure 2: Location of USB-C Port on Jetson Orin Development Kit

2. Set the board to recovery mode, as mentioned in below steps:
 - a. Press and hold the **Recovery** button of Jetson AGX Xavier™/ Orin™ development kit.
 - b. Press the **Power** button of Jetson AGX Xavier™/ Orin™ development kit.

The location of **Recovery** and **Power** buttons on the Jetson AGX Xavier™/ Orin™ development kit is shown in below figure.

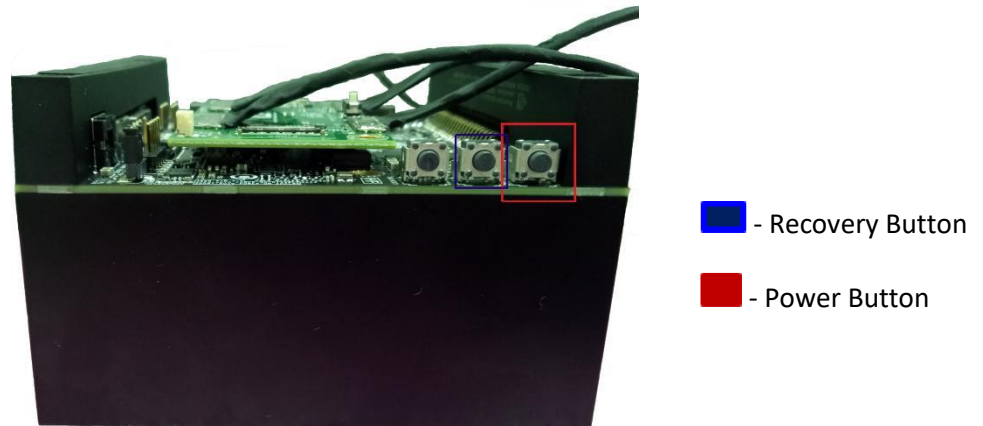


Figure 3: Location of Recovery and Power Buttons on Jetson AGX Xavier Development Kit



Figure 4: Location of Recovery and Power Buttons on Jetson Orin Development Kit

- c. Release both **Recovery** and **Power** buttons.

If the board is successfully changed to recovery mode, the Jetson AGX Xavier™/ Orin™ development kit will be enumerated as an USB device to the host PC.

3. Run the following command to verify whether the board is in recovery mode.

```
lsusb
```

The output message appears as shown below.

For Jetson AGX Xavier™

```
Bus 001 Device 102: ID 0955:7C18 NVidia Corp.
```

For Jetson Orin™

```
Bus 003 Device 006: ID 0955:7023 NVidia Corp.
```

4. Run the following flash.sh scripts to flash the Jetson AGX Xavier™ development kit from your host PC.

```
cd $L4T_DIR  
sudo ./flash.sh jetson-xavier mmcblk0p1
```

For Jetson Orin™,

```
sudo ./flash.sh jetson-agx-orin-devkit mmcblk0p1
```

Note: Now, the entire eMMC on the Jetson AGX Xavier™/Orin™ development kit and any files present on the device will be erased. It will take about 10-30 minutes to complete based on the host PC configuration.

5. Reboot the device.

Loading the Drivers

This section describes how to load the drivers, install the sample application and use the sample application with NileCAM81_CUOAGX.

The module drivers for NileCAM81_CUOAGX will be loaded automatically in the Jetson™ development kit during booting.

The steps to load the drivers are as follows:

1. Run the following command to check whether all the four cameras are initialized.

```
sudo dmesg | grep -i "Detected ar0821 sensor"
```

The output message appears as shown below.

```
Detected ar0821 sensor
```

The output message shows depends upon number of cameras are initialized properly.

2. Run the following command to check the presence of video node.

```
ls /dev/video*
```

The output message video (*) depends upon number cameras connected to the development kit and initialized properly. If no other cameras are connected to the Jetson AGX Xavier™ development kit. This video nodes can be utilized by any V4L2 application for viewing the camera preview.

The default login credentials of the Jetson AGX Xavier™ development kit are shown in below table.

Table 5: Default Login Credentials

Fields	Inputs
Username	Nvidia
Password	Nvidia

Installing the Sample Application

The NileCAM81_CUOAGX guvcviewer or ecam_tk1_guvcview is a simple GTK+ interface for capturing and viewing video from the devices supported on the Jetson AGX Orin™/Xavier™ development kit.

Using guvcviewer or ecam_tk1_guvcview application, you can perform the following:

- Enumerate and list all the video devices connected.
- Display properties of video renderer.

- Change resolution and color space or compression for video stream if different resolutions are supported by the device.
- Display currently configured values of preview.
- Capture the still images and set the path where still images will be saved.
- Display the average frame rate.

All the above listed properties can be configured by attractive and easy to use Graphical User Interface (GUI).

Please refer to the *e-CAM_TK1-GUVCView_Build_and_Install_Guide.pdf* for the procedure to build and install `ecam_tk1_guvcview` application on the Jetson AGX Orin™/Xavier™ development kit.

Using the Sample Applications with NileCAM81_CUOAGX

To use the `ecam_tk1_guvcview` application with NileCAM81_CUOAGX, please refer to the *NileCAM81_CUOAGX_Linux_App_User_Manual.pdf*

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

Is it possible to install the camera binaries without flashing the entire package?

Yes, please refer to the *Software quick Setup* section in *NileCAM81_CUOAGX_Getting_Started_Manual.pdf* to upgrade the modules, kernel image and device tree.

FAQ

1. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

What's Next?

After understanding how to setup the Jetson AGX Orin™/Xavier™ development kit using NileCAM81_CUOAGX MIPI camera, you can refer to the following documents to understand more about NileCAM81_CUOAGX.

- *NileCAM81_CUOAGX Release Notes*
- *NileCAM81_CUOAGX Release Package Manifest*
- *e-CAM_TK1-GUVView Build and Install Guide*
- *NileCAM81_CUOAGX Linux App User Manual*

Glossary

API: Application Programming Interface.

CMOS: Complementary Metal Oxide Semiconductor.

DTB: Device Tree Blob.

eMMC: Embedded Multi-Media Controller.

GIMP: GNU Image Manipulation Program.

GNU: GNU's Not Unix.

GTK: GIMP Toolkit.

GUI: Graphical User Interface.

MIPI: Mobile Industry Processor Interface.

L4T: Linux for Tegra.

Rootfs: Root Filesystem.

USB: Universal Serial Bus.

V4L2: Video4Linux2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	22-08-2022	Initial draft	Camera Dev Team
1.1	08-11-2022	Updated for Jetpack 5.0.2	Camera Dev Team
1.2	23-Jan-2023	Updated Sensor format from 1/2 to 1/1.7	Camera Dev Team
1.3	30-Jan-2023	Added dependency command	Camera Dev Team
1.4	22-Mar-2023	Updated to L4T35.2.1	Camera Dev Team
1.5	24-May-2023	Updated to L4T35.3.1	Camera Dev Team
1.6	19-OCT-2023	Updated to L4T35.4.1	Camera Dev Team
1.7	08-Jan-2024	Renamed the release package	Camera Dev Team